



Republic of Rwanda
Ministry of Education



RTB | RWANDA
TVET BOARD

COORDINATE MEASUREMENT

LSVCM402

Perform Coordinate Measurement

Competence

RQF Level: 4

Learning Hours



Credits: 8

Sector: Construction and Building Services

Trade: Land Surveying

Module Type: Specific

Curriculum: CBSBDC4002- TVET Certificate IV in Land Surveying

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Purpose statement	This module describes the knowledge, skills, and attitude required to perform coordinate measurement. It is designed for students pursuing RQF level 4 in Land surveying. By the end of this module, the students will be able to identify GNSS systems and total stations, prepare for coordinate measurement, measure coordinates and check errors in coordinate measurement.					
Learning assumed to be in place	Surveying tools and equipment, Surveying communication language, Linear distance measurement, Angular measurements, Levelling Surveys applications, applied mathematics, Surveying computation and adjustment.					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	50%
	Practical work:		70%		70%	
	Group work and presentation	30%				
	Individual work	40%				
			Summative Assessment		50%	

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
1. Identify GNSS and Total Stations	1.1. GNSS and Total station equipment is properly cased and uncased according to manufacturer's manual.
	1.2. GNSS types are properly identified according to their developers.
	1.3. Different GNSS segments are properly identified according to their types and functions.
	1.4. GNSS equipment is properly identified according to their manufacturers.
	1.5. GNSS coordinate measurement methods are properly identified according to their types.
	1.6. Basic structure components of Total stations are properly identified

	according to their functions.
	1.7. Total stations are properly classified based on the type of electromagnetic radiation used.
2. Prepare for coordinate measurement	2.1. PPE is properly worn & used according to the work to be done.
	2.2. Tools are appropriately selected in accordance to the work to be done.
	2.3. Materials are appropriately selected according to the work to be done.
	2.4. GNSS and Total station equipment are properly checked according to the manufacturer's specifications.
3. Measure coordinates	3.1. Total station is properly centred and levelled over the point according to manual specifications.
	3.2. Total station is properly configured according to the selected coordinate system and environmental conditions.
	3.3. Coordinates are properly measured according to Total station point positioning technique selected.
	3.4. The method of GNSS coordinate measurement is properly selected according to the work specifications.
	3.5. GNSS Receiver is properly centred and levelled according to the GNSS surveying method selected.
	3.6. GNSS Receiver is properly configured according to the selected coordinate system and environmental conditions.
	3.7. Coordinates are properly measured according to the GNSS surveying method selected.
4. Check errors in coordinate measurement	4.1. Error sources are properly identified in Total station measurement according to the survey method used.
	4.2. Error sources are properly identified in GNSS Measurement according to the survey method used.
	4.3. Error Misclosure is accurately computed according to the survey method used.

Intended Knowledge, Skills and Attitude

Knowledge	Skills	Attitude
✓ Describe basic principles of GNSS and Total station measurement ✓ Identify types of GNSS receiver and classification of Total stations ✓ Identify surveying materials, tools and equipment ✓ Identify methods of coordinate measurement ✓ Identify error in coordinate measurement	✓ Select and use material and tools for coordinate measurement ✓ Operate Total station and GNSS Receiver ✓ Apply total station and GNSS Receiver in coordinate measurement ✓ Apply safety precautions at workplace ✓ Apply principles of error computation ✓ Apply communication skills at workplace	✓ Respect time throughout the operation of Total station and DGNSS receiver ✓ Take care to error computation ✓ Show confidence while operating Total station and DGNSS receiver ✓ Demonstrate persistent, team working, flexibility, punctuality during the measurement of coordinates ✓ Solve problems related to total stations and DGNSS equipment malfunctioning ✓ Demonstrate risk awareness by putting on proper PPE

Course content

Learning outcomes	At the end of this module the student will be able to: 1. Identify GNSS systems and Total stations 2. Prepare for coordinate measurement 3. Measure coordinates 4. Check errors in coordinate measurement
Learning outcome 1: Identify GNSS and Total stations	Learning hours: 10

Indicative content

- **Casing and Uncasing GNSS receivers and Total stations**

- ✓ GNSS receiver
- ✓ Total stations

- **Identification of GNSS types**

- ✓ GPS (United States)
- ✓ GLONASS(Russia)
- ✓ GALILEO (European Union)
- ✓ Compass/ BeiDou (China)
- ✓ IRNSS/NavIC (Indian)
- ✓ QZSS (Japan)

- **Identification of GNSS segments**

- ✓ Types of GNSS Segments
 -  Space
 -  Control
 -  User
- ✓ Functions of GNSS segments

- **Identification of GNSS equipment**

- ✓ Types of GNSS equipment
 -  GNSS Receivers
 -  DGNSS Receivers
- ✓ Components of GNSS Receivers
 -  Receivers/Controllers
 -  Antenna
 -  Supporting software/Applications
 -  Accessories
- ✓ GNSS equipment manufacturers
 -  Trimble
 -  Leica
 -  Topcon,

 FOIF, etc

- **Identification of GNSS coordinate measurement methods**

- ✓ Absolute (autonomous) positioning
- ✓ Static positioning
- ✓ Relative positioning

- **Identification of Total station components**

- ✓ Main components
 -  Electronic theodolite
 -  EDM
 -  Microprocessor
- ✓ Parts of Total station
- ✓ Total station accessories

- **Classification of Total stations**

- ✓ Optical Total station
- ✓ Infrared Total station
- ✓ Microwave Total station

Resources required for the learning outcome

Equipment	Hand held GNSS Receivers, Differential GNSS Receivers, Total station, Computer, Projector, whiteboard/blackboard, PPE,
Materials	Overcoat, Field book, Pens, Marker pen, Pencil, First aid kit, Chalks
Tools	Plumb bob, Tape measures, Arrows, Prism pole, Reflector, Tripod, Communication Radios
Facilitation techniques	▪ Individual and group work ▪ Group discussion
Formative assessment methods /(CAT)	▪ Written assessment ▪ Oral presentation ▪ Performance assessment

Learning outcome 2: Prepare for coordinate measurement		Learning hours: 10	
Indicative content			
• Wearing PPE ✓ Surveyor’s Vest ✓ Safety boots ✓ Helmet • Selecting Tools ✓ Tripod & Plumb bob ✓ Reflector & Prism pole ✓ Communication radios ✓ Compass ✓ Measuring tape ✓ Ranging rods, Arrows ✓ Hammers, brushes • Selecting Materials ✓ Pegs ✓ Paint ✓ Nails ✓ Warning tape • Checking equipment ✓ Instrument connectivity ✓ Optical and laser plummet ✓ Battery ✓ Functioning of all other components			
Resources required for the learning outcome			
Equipment		Hand held GNSS Receiver, Differential GNSS Receiver, Total station, PPE, Computer, Projector, whiteboard/blackboard	
Materials		Field book, Pens, Marker pen, Pencil, Pegs, Nails, warning tape, Paint, Flags, First aid kit, Brushes, Cement, Crushed stones, Sand, Water	

Tools	Ranging rod, Plumb bob, Arrows, Prism pole, Reflector, Tripod, Compass, Communication Radios, Calculator, Tape measure, Hoe, Pick axes, Hammer, Panga, Safety cones.
Facilitation techniques	▪ Individual and group work ▪ Group discussion ▪ Practices
Formative assessment methods /(CAT)	▪ Written assessment ▪ Oral presentation ▪ Performance assessment

Learning outcome 3: Measure coordinates	Learning hours: 45
Indicative content	
• Centering and levelling the Total station ✓ Setup of tripod ✓ Centering Total station over station mark ✓ Levelling up Total station • Configuration of Total station ✓ Station and orientation ✓ Units of measurement ✓ Other parameters • Measuring coordinates with Total station ✓ Focussing Object ✓ Take measurement ✓ Record measurement • Selection of methods of GNSS coordinate measurements ✓ Absolute measurement ✓ Static positioning ✓ Relative positioning	

- **Centre and levelling of GNSS Receiver**
 - ✓ Setup of tripod
 - ✓ Centering GNSS base receiver over control point
 - ✓ Levelling up GNSS Receiver
- **Configuration of GNSS receiver**
 - ✓ Create and activate access account to geonet
 - ✓ Connect to CORS
 - ✓ Select datum and coordinate system
 - ✓ Set height of instrument
- **Measuring coordinates with GNSS Receiver**
 - ✓ Check of accuracy
 - ✓ Take measurement
 - ✓ Record measurement

Resources required for the learning outcome

Equipment	Hand held GNSS Receiver, Differential GNSS Receiver, Total station, PPE, computer, whiteboard / blackboard
Materials	Field book, Pens, Pegs, Nails, warning tape, Marker pen, Pencil, Paint, Flags, First aid kit
Tools	Hammer, Panga, ranging rod, Plumb bob, Tape measures, Arrows, Prism pole, Compass, Calculator, Reflector, Tripod, Communication Radios, Hoe, Pick axes, Safety cones.
Facilitation techniques	▪ Demonstration ▪ Individual and group work ▪ Group discussion ▪ Field practices
Formative assessment methods /(CAT)	▪ Written assessment ▪ Oral presentation ▪ Performance assessment

Learning outcome 4: Check errors in coordinate measurement		Learning hours: 15	
Indicative content			
• Identification of error sources in Total station measurement ✓ Instrument errors ✓ Personal errors ✓ Atmospheric errors • Identification error sources in GNSS receiver ✓ Atmospheric condition ✓ Clock error ✓ Ephemeris (orbital path) data error ✓ Multi-path effects • Computation of misclosure in coordinates measurements ✓ Angular misclosure ✓ Linear misclosure ✓ Adjust error of closure			
Resources required for the learning outcome			
Equipment		Hand held GNSS Receiver, Differential GNSS Receiver, Total station, PPE, computer, whiteboard/blackboard, projector	
Materials		Field book, Pens, Pegs, Nails, warning tape, Marker pen, Pencil, Paint, Safety cones, Flags, First aid kit.	
Tools		Hammer, Panga, ranging rod, Plumb bob, Tape measures, Arrows, Prism pole, Compass, Calculator, Reflector, Tripod, Communication Radios, Hoe, Pick axes.	
Facilitation techniques		▪ Demonstration ▪ Individual and group work ▪ Group discussion	
Formative assessment methods /(CAT)		▪ Written assessment ▪ Oral presentation ▪ Performance assessment	

Integrated/Summative assessment

Integrated situation

Land Solution Ltd is requested by Mr KAGABO Aimable, a resident of Umucyo Village, Rusera cell, Kabarondo Sector in Kayonza District to Provide data for the production of topographic map of his parcel where he want to construct a primary school. The plot has the area of 1ha approximately. As a surveyor assistant employed by Land Solution Ltd, you have been requested to collect all required data to produce the topographic map of that area within eight (8) hours.

The control points should be established on the site by close traverse referring to the existing control points already located at approximately 1km from the surveying area and should be corrected for closing error. Data (coordinates) should be measured using either Total station or DGNSS Receiver.

Tools	▪ Hammer, Panga, ranging rod, Plumb bob, Tape measure, Arrows, Prisms, Calculator, Reflector, Tripod, Communication Radios
Equipment	▪ Differential GNSS Receiver, Total station, PPE, computer
Materials/ Consumables	▪ Overcoat, Overall, Field book, Pens, Pegs, warning tape, Marker pen, Pencil, Safety cones,

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
1. Identify GNSS systems and total stations 10%	1.1 GNSS and total station equipment is properly cased and uncased	1. GNSS equipment is appropriately cased and uncased			3
		2. Total station equipment is appropriately cased and uncased			3
	1.2 GNSS types are properly identified				
	1.3 Different GNSS segments are properly identified	1. Satellite constellation is well checked			4
	1.4 GNSS equipment is properly identified	2. GNSS Receivers identified			

	1.5 Basic structure components of total stations identified properly	1 Components of total station identified			
	1.6 Total stations properly classified	2 Classification of total station identified			
2. Prepare for coordinate measurement 20%	2.1. PPE is properly worn & used	1 PPE is appropriately worn			2
	2.2. Tools, materials and equipment are appropriately selected	1 Tools are well selected			2
		2 Materials are well selected			2
		3 Equipment are well selected			3
	2.3. GNSS and Total station equipment are properly checked	1 Instrument connectivity is accurately checked			3
		2 Optical and laser plummet is accurately checked			3
		3 Battery is checked			2
		4 Functioning of other components is properly checked			3
3. Measure coordinates 50%	3.1. Total station is properly centred and levelled	1. Total station is properly centred			3
		2. Total station is properly levelled			3
	3.2. Total station is properly configured	1. Station (s) and orientation (s) are well set			3
		2. Units of measurement are well set			2
		3. Other parameters are well set			3
	3.3. Coordinates are properly measured	1. Coordinates are accurately measured			5

		2. Coordinates are properly recorded			3
	3.4. The method of GNSS coordinate measurement is properly selected	1. The method of GNSS coordinate measurement is selected			3
	3.5. GNSS Receiver is properly centred and levelled	1. GNSS Receiver is centred			3
		2. GNSS Receiver is levelled			3
	3.6. GNSS Receiver is properly configure	1 Creation and activation of access account to geonet is well done			3
		2 Connecting to CORS is well done			3
		3 Selection of datum and coordinate system is well done			3
		4 Setting of instrument height is well done			2
	3.7. GNSS surveying coordinates are properly measured	1 GNSS surveying coordinates are accurately measured			5
		2 GNSS surveying coordinates are properly recorded			3
4. Check errors in coordinate measurement 20%	4.1. Total station measurement errors are properly identified	1. Total station measurement errors are properly checked			2.5
	4.2. Errors are properly identified in GNSS Measurement	1. GNSS measurement errors are properly checked			2.5

	4.3. Error misclosure is accurately computed	1. Angular misclosure is properly computed			5
		2. Linear misclosure is properly computed			5
		3. Error of closure is accurately adjusted			5
Total marks		100%			
Percentage Weightage		100%			
Minimum Passing line % (Aggregate): 70%					

References

1. Lewis, M. J. T. (23 April 2001). *Surveying Instruments of Greece and Rome*. Cambridge University Press. ISBN 9780521792974. Retrieved 30 August 2012.
2. Ahun, Jerry. "Electronic Distance Measurement". Jerrymahun.com. Archived from the original on 29 July 2014. Retrieved 20 July 2014.
3. Nathanson Kissam, 1988, *Surveying Practice*, 4th Edition,